WEST BENGAL JEE 2026

MATHEMATICS MOCK TEST QUESTION

General Instructions

- This paper contains 75 questions divided into three categories.
- **Category-I:** 1 Mark each. Only one option correct. Negative marking: -0.25.
- **Category-II:** 2 Marks each. Only one option correct. Negative marking: -0.5.
- **Category-III:** 2 Marks each. One or more options may be correct. No negative marking.
- Partial marks are awarded for partially correct answers in Category-III.

Category: I (Q: 1 to 50)

1 Mark Each. Only one correct option.

1. Let f(x)=sqrt(log10(x^2)). The set of all values of x for which f(x) is real is

(A) $[1, \infty]$

(B) $[-1, 1]$

(C) $[-\infty, -1]$

(D) $[-\infty, -1] \cup [1, \infty]$

Ans: (D)

2. If log_cos(x)tan(x) + log_sin(x)cot(x) = 0, the general solution for x is

(A) $n\pi + \pi/4$

(B) $2n\pi + \pi/4$

(C) $2n\pi - 3\pi/4$

(D) $n\pi - \pi/4$

Ans: (B)

3. If x(x^2+1), -x^2/2, 6 are consecutive terms of an A.P., the next two are

(A) 14, 6

(B) -2, -10

(C) 14, 22

(D) -2, 6

Ans: (C)

4. If a,b,c are real, $a+b+c<0$ and $ax^2+bx+c=0$ has imaginary roots, then

(A) $a>0, c>0$

(B) $a<0, c>0$

(C) $a>0, c<0$

(D) $a<0, c<0$

Ans: (D)

5. If $x^2 + 9y^2 + 25z^2 = xyz(15/x + 5/y + 3/z)$ then x, y, z are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none

Ans: (C)

6. Solution of the inequation $|z-4| < |z-2|$ is

(A) Re $z > 0$

(B) Re $z < 0$

(C) Re $z > 3$

(D) Re $z < 3$

Ans: (C)

7. The argument of the complex number $z = 1 + i \tan(2\pi/5)$ is

(A) $-2\pi/5$

(B) $2\pi/5$

(C) $-3\pi/5$

(D) $3\pi/5$

Ans: (A)

8. How many words containing 3 consonants and 2 vowels can be formed from 10 consonants and 4 vowels?

(A) 10C3 * 4C2 * 5!

(B) 10C3 * 5!

(C) 10C3 * 4C3

(D) 10C3

Ans: (A)

9. If $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$, then $g\{f(x)\}$ is invertible if

(A) $-\pi/2 \le x \le 0$

(B) $-\pi/2 \le x \le \pi$

(C) $0 \le x \le \pi/2$

(D) $-\pi/4 \le x \le \pi/4$

Ans: (D)

10. The value of $\lim_{x \to 0} (1+3x)^{(x+2)/x}$ is

(A) e^3

(B) e^6

(C) e^5

(D) e

Ans: (B)

11. If $2^x + 2^y = 2^{(x+y)}$, the value of dy/dx at $(1,1)$ is

(A) 0

(B) -1

(C) 1

(D) 2

Ans: (B)

12. Let $f(x) = x(x-1)(x-2)$ on $[0,4]$. The point $x=c$ satisfying MVT is

(A) 0

(B) $c>3$

(C) 0

(D) 1

Ans: (D)

13. Integral of $x^5 / \text{sqrt}(1+x^3)$ dx is

(A) $2/3[(1+x^3)^{(3/2)}/3 - \text{sqrt}(1+x^3)] + c$

(B) $1/3[(1+x^3)^{(3/2)}/3 - \text{sqrt}(1+x^3)] + c$

(C) $2/3[(1+x^3)^{(3/2)}/3 + \text{sqrt}(1+x^3)] + c$

(D) $1/3[(1+x^3)^{(3/2)}/3 + \text{sqrt}(1+x^3)] + c$

Ans: (A)

14. If Integral $(4e^x+6e^{-x})/(9e^x-4e^{-x})$ dx = $Ax + B \log(9e^{2x} - 4) + C$, then A, B, C are

(A) $A=5/2$, $B=31/36$, $C=-3/2 \log 3 + c$

(B) $A=-3/c^2$, $B=35/36$, $C=-3/2 \log 3 + c$

(C) $A=5/2$, $B=31/36$, $C=3/2 \log 3 + c$

(D) $A=-5/2$, $B=31/36$, $C=3/2 \log 3 + c$

Ans: (B)

15. Integral of $\sin x / \sin(x-\alpha)$ dx is

(A) $x \cos \alpha + \sin \alpha \log \sin(x-\alpha) + c$

(B) $x \cos \alpha - \sin \alpha \log \sin(x-\alpha) + c$

(C) $x \cos \alpha + \sin \alpha \log \sin(x+\alpha) + c$

(D) $x \cos \alpha - \sin \alpha \log \sin(x+\alpha) + c$

Ans: (A)

16. Integral from $-\pi/2$ to $\pi/2$ of $[f(x)+f(-x)][g(x)-g(-x)]$ dx (f, g continuous) is

(A) 0

(B) 1

(C) -1

(D) 2

Ans: (A)

17. If $f(x) = \{2x^2+1, x \le 1; 4x^2-1, x>1\}$, then integral from 0 to 2 of $f(x)$ dx is

(A) 47/3

(B) 50/3

(C) 1/3

(D) 47/2

Ans: (A)

18. The order of differential equation: $2x^2 y'' - 3y' + y = 0$ is

(A) 2

(B) 1

(C) 0

(D) undefined

Ans: (A)

19. Number of arbitrary constants in general solution of 4th order diff eq is

(A) 0

(B) 2

(C) 3

(D) 4

Ans: (D)

20. General solution of $e^x dy + (ye^x + 2x) dx = 0$ is

(A) $xe^x + x^2 = c$

(B) $xe^y + y^2 = c$

(C) $ye^x + x^2 = c$

(D) $xe^x - x^2 = c$

Ans: (C)

21. Simplified form of $\cos^{-1}(4x^3 - 3x)$ is

(A) $3 \sin^{-1} x$

(B) $3 \cos^{-1} x$

(C) $\pi - 3 \sin^{-1} x$

(D) $\pi - 3 \cos^{-1} x$

Ans: (B)

22. For triangle ABC, value of b^2 sin 2C + c^2 sin 2B is

(A) Δ

(B) 2Δ

(C) 3Δ

(D) 4Δ

Ans: (D)

23. If tan α = a/(a+1) and tan β = 1/(2a+1), then α+β is

(A) π

(B) π/2

(C) π/3

(D) π/4

Ans: (D)

24. For triangle ABC, value of b^2 sin 2C + c^2 sin 2B is

(A) Δ

(B) 2Δ

(C) 3Δ

(D) 4Δ

Ans: (D)

25. If sin A + sin B = 2 then value of sin(A+B) is

(A) 2

(B) 0

(C) 1

(D) -1

Ans: (B)

26. If tan 25 = a, then (tan 155 - tan 115) / (1 + tan 155 tan 115) is

(A) 2a/(1-a^2)

(B) 2a/(1+a^2)

(C) (1-a^2)/2a

(D) (1+a^2)/2a

Ans: (C)

27. If length of conjugate axis > transverse axis in hyperbola, eccentricity e is

(A) $\sqrt{2}$

(B) $>\sqrt{2}$

(C)

(D) $1/\sqrt{2}$

Ans: (B)

28. Number of intersecting points of $4x^2+9y^2=1$ and $4x^2+y^2=4$ is

(A) 1

(B) 2

(C) 3

(D) 0

Ans: (D)

29. Foci of hyperbola $x^2-y^2+1=0$ form diameter of circle. Circle equation is

(A) $x^2+y^2=4$

(B) $x^2+y^2=\sqrt{2}$

(C) $x^2+y^2=2$

(D) $x^2+y^2=2\sqrt{2}$

Ans: (C)

30. Q is image of P(1,5) about y=x. R is image of Q about y=-x. Circumcentre of PQR is

(A) (5,1)

(B) (-5,1)

(C) (1,-5)

(D) (0,0)

Ans: (D)

31. Foot of perpendicular from (1,8,4) on line joining (0,-11,4) and (2,-3,1) is

(A) (4,5,2)

(B) (-4,5,2)

(C) (4,-5,2)

(D) (4,5,-2)

Ans: (D)

32. Intersection of $2ax+4ay+c=0$ and $7bx+3by-d=0$ is in 4th quadrant and equidistant from axes. ad:bc is

(A) 2:3

(B) 2:1

(C) 1:1

(D) 3:2

Ans: (B)

33. Let P(n): $n^3+n=3m$. Which is true?

(A) P(1)

(B) P(2)

(C) P(3)

(D) P(4)

Ans: (C)

34. If sum of 5th and 6th terms of $(a-b)^n$ is 0, a/b is

(A) 5/(n-4)

(B) 1/5(n-4)

(C) (n-5)/6

(D) (n-4)/5

Ans: (D)

35. The coefficient of x^3 in expansion of 3^x is

(A) $(\log_e 3)^3 / 6$

(B) $(\log_e 3)^3 / 3$

(C) $\log_e 3 / 6$

(D) $(\log_c 3)^3 / 3$

Ans: (A)

36. If A and B are symmetric matrices then

(A) AB+BA is symmetric

(B) AB-BA is symmetric

(C) AB-BA is skew-symmetric

(D) AB+BA is skew-symmetric

Ans: (C)

37. If A=[3 5; 2 0] and B=[1 17; 0 -10] then |AB| is

(A) 80

(B) 100

(C) 110

(D) 92

Ans: (B)

38. Roots of det[[x,2,1],[2,5,x],[-1,2,x]] are

(A) -3,1

(B) 3,-1

(C) 3,1

(D) -3,-1

Ans: (B)

39. If A=[1 1 3; 5 2 6; -2 -1 -3], A is

(A) involutary matrix

(B) nilpotent matrix of index 2

(C) nilpotent matrix of index 3

(D) rectangular matrix

Ans: (C)

40. Balls from 3 boxes (3W,1B), (2W,2B), (1W,3B). Prob of 2W 1B is

(A) 13/32

(B) 1/4

(C) 1/32

(D) 3/16

Ans: (A)

41. Independent events probabilities 3/10 and 2/5. Prob of exactly one happening is

(A) 23/50

(B) 1/2

(C) 31/50

(D) None

Ans: (A)

42. Set theory: (A U B)' U (A' n B) is

(A) A U B

(B) A'

(C) A

(D) None

Ans: (B)

43. Variance of 0, 10, 20, 30, 40, 50 is

(A) 291.67

(B) 290

(C) 230

(D) 12

Ans: (A)

44. A={1,2,3}, R={(1,2),(2,1)}. R is

(A) reflexive if (1,1) added

(B) symmetric if (2,3) added

(C) transitive if (1,1) added

(D) symmetric if (3,2) added

Ans: (C)

45. r on N x N: (a,b)r(c,d) => a+d=b+c. r is

(A) Reflexive only

(B) Symmetric only

(C) Transitive only

(D) Equivalence relation

Ans: (D)

46. Curve ax+by+xy=0 makes angle tan^-1 2 with X axis at (1,1). a, b are

(A) 1, 2

(B) 2, 1

(C) -2, 1

(D) 1, -2

Ans: (D)

47. Length of normal at theta=pi/2 on x=a(e+sin theta), y=a(1-cos theta)

(A) 2a

(B) a/2

(C) a sqrt(2)

(D) a/sqrt(2)

Ans: (C)

48. Function f(x) = cos x - 2kx is decreasing when

(A) k<2

(B) k>2

(C) k>1/2

(D) k<1/2

Ans: (C)

49. If phi(x)=f'(x)+f'(2a-x) and f''(x)>0 on [0,a], then phi(x) is

(A) increasing in [0,a]

(B) increasing in [0,2a]

(C) decreasing in [0,a]

(D) increasing in (0,a)

Ans: (C)

50. Area of {(x, y): |x| >= y >= x^2} is

(A) 2/3

(B) 1/3

(C) 1/6

(D) None

Ans: (B)

Category: II (Q: 51 to 65)

2 Marks Each. Only one correct option.

51. If x, y, z are positive in A.P. then

(A) $(x+y)/(2y-x) + (y+z)/(2y-z) \geq 4$

(B) $(x+y)/(2y-x) + (y+z)/(2y-z) \geq 2$

(C) $y^2 \leq zx$

(D) $(x+y)/(2y-x) + (y+z)/(2y-z) \leq 2$

Ans: (A)

52. Quadratic eq whose roots are A.M. and H.M. of two numbers a, b is

(A) $Ax^2 - (A^2+H^2)x + AH^2 = 0$

(B) $Ax^2 - (A^2+G^2)x + AG^2 = 0$

(C) $Hx^2 - (H^2+G^2)x + HG^2 = 0$

(D) none

Ans: (B)

53. If $|z1|=12$ and $|z2-3-4i|=5$, minimum of $|z1-z2|$ is

(A) 7

(B) 0

(C) 2

(D) 17

Ans: (B)

54. Words from SUCCESS so that two C are together but no two S are together

(A) 24

(B) 42

(C) 43

(D) 23

Ans: (A)

55. Set of points where $f(x)=x|x|$ is twice differentiable is

(A) all x in R

(B) x in $R-\{0\}$

(C) x in $R-\{0,1\}$

(D) x in $R-\{1\}$

Ans: (B)

56. If $e^{f(x)} = (10+x)/(10-x)$ and $f(x)=kf(200x/(100+x^2))$, then k is

(A) $1/2$

(B) $4/5$

(C) $7/10$

(D) $2/3$

Ans: (A)

57. If $f(x)=2x+3$, $g(x)=x^2+7$, x values such that $(g \circ f)(x)=8$ are

(A) 1, -2

(B) -1, 2

(C) -1, -2

(D) 1, 2

Ans: (C)

58. General solution of $(1+e^{x/y})dx + (1-x/y)e^{x/y}dy = 0$ is

(A) $x - ye^{x/y} = c$

(B) $y - ye^{x/y} = c$

(C) $x + ye^{x/y} = c$

(D) $y + ye^{x/y} = c$

Ans: (C)

59. Values of x satisfying $\tan^{-1}((x-1)/(x-2)) + \tan^{-1}((x+1)/(x+2)) = \pi/4$ are

(A) $\pm 1/\sqrt{2}$

(B) $\pm \sqrt{2}$

(C) $\pm 1/2$

(D) ± 2

Ans: (A)

60. If $P = 1/2 \sin^2 \theta + 1/3 \cos^2 \theta$, then

(A) $1/3 \le P \le 1/2$

(B) $P \le 1/2$

(C) $2 \le P \le 3$

(D) $P \le 1/3$

Ans: (A)

61. Length of chord of parabola y^2=4ax passing through vertex making acute angle theta with axis

(A) +- 4a cot theta cosec theta

(B) 4a cot theta cosec theta

(C) -4a cot theta cosec theta

(D) 4a cosec^2 theta

Ans: (B)

62. l, m, n are pth, qth, rth terms of G.P. Value of det[[log l, p, 1], [log m, q, 1], [log n, r, 1]] is

(A) -1

(B) 2

(C) 1

(D) 0

Ans: (D)

63. Coefficient of b^10 a^7 c^3 in expansion of (ab + bc + ca)^10

(A) 100

(B) 110

(C) 120

(D) 130

Ans: (C)

64. Area of region bounded by y=x^2+2, y=-x, x=0, x=1 is

(A) 13/6

(B) 15/8

(C) 17/6

(D) none

Ans: (C)

65. If independent events P(A)=1/5, P(A U B)=7/10, then P(B_bar) is

(A) 3/8

(B) 2/7

(C) 7/9

(D) none

Ans: (A)

Category: III (Q: 66 to 75)

2 Marks Each. Multiple correct options.

66. If x^2 + 4y^2 = 12xy, x,y in [1,4], which are true?

(A) Max value of log2(x+2y) is 4

(B) Min value of log2(x+2y) is 3

(C) log2(x+2y) in [2,4]

(D) 2 integral pairs (x,y) such that log2(x+2y)=3

Ans: (A, C, D)

67. Let f(x)=x^2(x+2)+x+3. Then

(A) f(-3-k)>0 and f(-2+k)<0

(B) f(-3-k)<0 and f(-2+k)>0

(C) f(x)=0 has root a s.t. [a]+3=0

(D) f(x)=0 has root a s.t. (a)+2=0

Ans: (B, C, D)

68. If |(2z-i)/(z+1)|=m represents a circle, m can be

(A) 1/2

(B) 1

(C) 2

(D) 3

Ans: (A, B, D)

69. Number of ways to select 3 numbers in A.P from 1,2,...,n

(A) $((n-1)/2)^2$ if even

(B) $n(n-1)/4$ if even

(C) $((n-1)^2)/4$ if odd

(D) none

Ans: (B, C)

70. If $f(x) = \{x^2, x>=0; -x^2, x<0\}$

(A) Not differentiable at 0

(B) Discontinuous at 0

(C) Differentiable at 0

(D) Continuous at 0

Ans: (A, D)

71. Let I_n = integral(0 to 1) $x^n \tan^{-1} x$ dx. If $a_n I_{n+2} + b_n I_n = c_n$

(A) a_1, a_2, a_3 in G.P.

(B) b_1, b_2, b_3 in A.P.

(C) c_1, c_2, c_3 in H.P.

(D) a_1, a_2, a_3 in A.P.

Ans: (B, D)

72. Hyperbola $16x^2 - 3y^2 - 32x - 12y = 44$. Then

(A) Transverse axis is $2\sqrt{3}$

(B) Latus rectum is $32\sqrt{3}$

(C) Eccentricity is $\sqrt{19/3}$

(D) Directrix is $x=\sqrt{19/3}$

Ans: (A, B, C)

73. Line through (a, b) and (-a, -b) passes through

(A) (1,1)

(B) (3a, -2b)

(C) (a^2, ab)

(D) (a, b)

Ans: (C, D)

74. Delta(x) = det[[sin x/2, 1, 1], [1, sin x/2, -sin x/2], [-sin x/2, 1, -1]] on [0,pi]

(A) Max at x=pi

(B) Min at x=0

(C) Max at 0, Min at pi

(D) Range is [2,4]

Ans: (A, B, D)

75. Vectors a=ai+j+2k, b=i+aj-k, c=2i-j+ak are coplanar if

(A) a=-2

(B) a=sqrt(3)+1

(C) a=1-sqrt(3)

(D) a=2

Ans: (A, B, C)